

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM

(An Institute of National Importance under Ministry of Education, Govt. of India, governed by an Act of Parliament)

The Institute

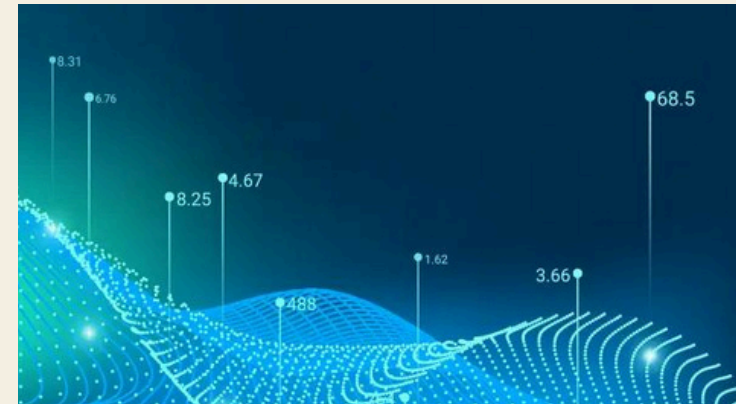
The Indian Institute of Information Technology, Kottayam is one among the IITs that has been established as "Institutions of National Importance" by the Ministry of Education, Govt. of India under the ambit of IIT (PPP) Act 2017. The institute offers 4 year B.Tech and B.Tech(Hons) program with specialized courses covering all major research domains of Computer Science and Electronics and Communication Engineering. The students are admitted against the JoSAA/CSAB counselling against the JEE main rank list. The institute also offers M.Tech in CSE and Ph.D. programs in CSE, ECE and Mathematics.

The Department

The Department of Computational Science and Humanities was formally established in 2019 with an aim to develop mathematically sound young minds for solving industry related problems arising from different industries and related areas. It offers core courses in Mathematics/ Statistics/Humanities to all B.Tech., M.Tech, and Ph.D. programs. This department takes pride in offering an exclusive doctoral program in Mathematics, meticulously crafted to meet the requirements of highly motivated and aspiring graduates holding a degree in this field.

Department of Computational Sciences
and Humanities, IIIT Kottayam
Presents

Bridging the Gap: Computational Math and AI (Hybrid Mode)



**Two-day FDP cum Workshop
June 27-28, 2024**

Contact Us

Program Coordinators:

- **Dr. Murugan D - 04822202249**,
Asst. Prof., Dept. of Computational Science
and Humanities, IIIT Kottayam
murugand@iiitkottayam.ac.in
- **Dr. Riyasudheen TK - 04822202256**,
Asst. Prof., Dept. of Computational Science
and Humanities, IIIT Kottayam
riyasudheen@iiitkottayam.ac.in

Program Convener:

- **Dr. Dhanyamol MV - 04822202162**
Asst. Prof. & Head, Dept. of Computational
Science and Humanities, IIIT Kottayam
dhanya@iiitkottayam.ac.in

Indian Institute of Information Technology, Kottayam
(IIITK), Valavoor - Chakkampuzha Rd, Valavoor, Kerala
686635, Phone: 04822 202 100, www.iiitkottayam.ac.in



About Workshop

"Bridging the Gap: Computational Math and AI" is a dynamic workshop aimed at raising students and the research community with essential programming skills and numerical packages. Participants will be involved in hands-on experiences, implementing computational methods to solve real-world problems. This workshop serves as a unique platform for exploring the intersection of computational mathematics and cutting-edge AI/ML, data science, and engineering domains, highlighting the interdisciplinary nature of these fields. By immersing themselves in practical applications, attendees will gain valuable insights into the latest AI advancements in computational mathematics, particularly in fields like computational finance and computational fluid dynamics, bringing up a deeper understanding and appreciation for the necessary relationship between mathematics and artificial intelligence.

Workshop Objectives

- Upskill the students and research community with programming languages and other numerical packages.
- Hands-on Experience of the implementing AI- assisted computational methods.
- Experience the intersection of computational mathematics, Artificial Intelligence (AI), Machine Learning (ML), data science, and engineering domains, exploring the interdisciplinary nature.
- Raise awareness of the latest AI advancements in fields such as computational finance and computational fluid dynamics.

Registration

Target Audience: Faculty, PhD scholars and UG/PG students.

Registration deadline: 20-06-2024

Registration Link:

<https://forms.gle/S4fqzcwbdahsXyd6>

(Seats will be assigned on a first-come, first-served basis due to limited availability)

***Fee payment details are provided within the registration process.**

Registration Fees

Please note : Registration fee is non refundable.

Participant Type	Online	Offline
PhD/PG/UG	500/-	1000/-
Faculty	1000/-	1500/-

***Only working lunch and tea will be provided to the offline participants.**

Accommodation fee Details

Single room : ₹200/- per day

Shared room: ₹100/- per day

Provided upon request via mail and mentioned the same in the registration link.]

Patrons

- **Prof. (Dr.) Rajiv Dharaskar, Director, IIIT Kottayam.**
- **Dr. M.Radhakrishnan, Registrar, IIIT Kottayam.**
- **Prof. (Dr.) P. Mohanan, Prof. In-Charge (Academics), IIIT Kottayam.**

Resource Persons

- Dr. Satyanda Panda, NIT Calicut
- Dr. E Natarajan, IIST Trivandrum
- Dr. Bakkyaraj, IIIT Kottayam
- Dr. Murugan D, IIIT Kottayam
- Dr. Jisha CR, UNIST, Republic of Korea
- Dr. Riyasudheen TK, IIIT Kottayam

Workshop Contents

- **Introduction to Physics Informed Neural Networks (PINN) and its significance in AI-assisted PDE solvers.**
- **Efficient AI-solvers for linear system of equations.**
- **Entropy stable schemes for compressible Navier-stocks equations using feed forward neural network.**
- **Introduction to Differential Quadrature Methods (DQM) in combination with Finite Differences (FDM).**
- **Introduction to Multi-grid Methods.**

Register Now:

